

Hardware Determinism over Statistical Gradients:

A GPU-Accelerated Boolean Attack on the Final U-534
Enigma M4 Ciphertext (P1030680)

Digital Defiance
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Abstract

For decades, message P1030680 has remained the sole unbroken Enigma M4 transmission from the May 1, 1945 intercept of German submarine U-534. Modern attempts to decrypt this 72-letter ciphertext have historically relied on statistical hill-climbing and distributed computing. This report shows that purely statistical attacks fail against short M4 ciphertexts due to plugboard overfitting, documents a mid-message middle-rotor turnover flaw in whole-message trigram scoring, and describes a software Welchman diagonal board on Apple Silicon Metal (40 million machine settings per second) with a mandatory ≤ 10 -plug completion kill chain. We record graded controls (known-message break), archival eliminations, clean negatives under catalog and operational cribs, a quarantine path for soft near-misses, and a parallel TensorLUT continuous-discrete compiler track. **We do not claim a decrypt of P1030680.** Campaign fitness is cleartext Metal batch, not HELUT's encrypted torus path (documented separately in `paper/helut.tex`).

1 The Failure of Statistical Gradients on Short Ciphertexts

When attempting to break an Enigma ciphertext, modern cryptographic approaches default to statistical optimization, such as Index of Coincidence (IC), n -gram fitness, and genetic algorithms. However, empirical testing against P1030680 proves that statistics cannot *search* a 72-letter Enigma space; they can only *rank* a small list of survivors.

The Enigma plugboard (the *Steckerbrett*) offers roughly 47 bits of freedom. On a short message, this immense freedom causes statistical models to overfit. Algorithms will output high-scoring strings of letters that closely resemble German syntax, but these strings are mathematical hallucinations. Control tests on a known, solved message (P1030684) confirmed that purely statistical engines favor these false positives over the true key. Consequently, any successful attack on short M4 intercepts must abandon statistical guessing in favor of absolute Boolean logic.

2 Emulating the Welchman Diagonal Board in GPU Shaders

To execute a deterministic attack, the hardware architecture of the 1945 Enigma must be perfectly mapped to modern hardware. Using Homomorphic Edge Look-Up Tensors (HELUT), we engineered a digital Bombe inside Apple Silicon Metal shaders.

This software-defined FPGA evaluates Gordon Welchman's diagonal board graph reductions at a rate of 40 million machine settings per second. The engine utilizes Turing-shaped reductions based on the physical reality of the short-message M4:

- Because the far-left Enigma rotor (the Greek wheel) never steps during a short message, its ring can be digitally pinned.

- Because the left rotor’s notch drives nothing, its ring can also be pinned.
- Together, this physical constraint collapses the total search space by a massive factor of 676.

The mathematical soundness of this reduction was verified in a blind control run. Given a 27-letter crib from a known message (P1030684) and no other parameters, the GPU pipeline evaluated 16 billion settings and extracted the true key, exact shell, and all ten historical plugboard cables in just 361 seconds (claim C2). The archived log (`logs/control-p1030684-rings.log`) reads 427 s: 361 s was the first run, and the log was overwritten by a re-run after head scoring was added. Both figures are recorded in `BREAK_P1030680.md`; the re-run is the reproducible one.

3 Physical Constraints vs. Mathematical Ghosts

Translating mathematical graph theory into physical 1945 hardware exposed a significant vulnerability in pure Boolean reductions. Certain crib wedges produce valid Welchman states that are mathematically flawless but physically impossible.

We classify these false positives as “ghosts”. In a trial run, a crib of `UUUVIRSIBENNULEINS` at offset 0 produced 12 mathematical survivors out of the Welchman board. However, when mapping the remaining letters, none of these 12 settings could be completed within the physical limit of exactly 10 plugboard cables. Forensics revealed these were “split menus,” where connected components within the graph admitted zero valid starting seeds.

To prevent these ghosts from overwhelming the CPU, we built a SAT-completion kill chain inline as the GPU drains. This ensures that only mathematically valid states that *also* fit physical hardware constraints survive the hardware pipeline (claim C3).

4 The Mid-Message Turnover Scoring Flaw

Our research uncovered a systemic flaw in how distributed computing projects historically score Enigma decrypts, likely explaining the decades-long failure to crack P1030680.

Standard algorithmic sweeps utilize a linguistic scanner to score the full 72-letter plaintext. However, if the true key clicks the middle rotor over *after* the crib phrase but *before* the end of the message, the plaintext will be perfectly decrypted across the crib span, but turn to garbage for the remainder of the message.

Because traditional trigram discriminators score the *entire* message, the garbled tail drags the overall score down to the midpoint between German and random noise. This causes the engine to discard the true key, actively rejecting the correct answer.

To rectify this, we designed a **windowed discriminator score**. This metric evaluates the readable head of a decrypt (requiring at least 16 non-crib letters) independently from the whole message, passing the key if either the full message or the head meets the -3.600 threshold. Applying this fix revealed 2,219 physically valid ten-plug boards that had been previously rejected by the flawed whole-message scoring (claim C12).

5 The M-Thetis / Potsdam Register Mismatch

Cryptanalysis is limited by the quality of the plaintext hypotheses (cribs) used as wedges. Archival auditing of the May 1, 1945 U-534 corpus highlighted a severe historical blindspot regarding the target message.

The scraped corpus contains 48 broken M4 messages. Crucially, 31 originate from the Potsdam net, 16 from a second net, and 1 from a third. P1030680 is the *only* message originating from the M-Thetis network.

Because Bletchley Park never bothered to work the M-Thetis training net, and no other Thetis messages survive in the corpus, all 100 historical cribs used to attack P1030680 were register language borrowed from other operational networks. If Thetis operators opened their 72-letter messages with callsigns or key groups before the body—unlike Potsdam operators—the exact crib placements used in modern attacks are fundamentally the wrong shape. The resilience of P1030680 is therefore likely a result of mismatched operational doctrine rather than the cryptographic strength of the cipher itself.

Potsdam and Plaipe daily keys for 1 May were eliminated by exhaustion against P1030680 (claim C11): all 26^4 message keys land at the noise floor.

6 Graded Campaign Status (August 2026)

The operational ledger is `BREAK_P1030680.md`; the public claim freeze is `directives/claim-sheet.md`. Table 1 summarizes what is *proven*, what is *eliminated*, and what remains open. **None of these rows is a decrypt of P1030680** (non-claim N5).

Result	Status
Welchman blind control (P1030684)	Break in 361 s first run / 427 s archived re-run; all 10 plugs
Stochastic KPA datapath	Control oracle + near-miss PASS; blind stecker $\sim 22/72$
Training / collapse / weather / RIGA priors	0 survivors under 80% rigor (ceiling $\sim 60\text{--}69\%$ = coincidence)
Exact ≥ 16 catalog $\times$ rings AAAA	Clean negative
Curated exact / fuzzed top-40 $\times$ right rings	Clean negative
UEBUNG / Thetis-register arms	Clean negative (register rings parked at 13/73)
Soft-tail UEBUNG quarantine escalate	55 seeds; Hybrid 0 survivors
Regenbogen / Hannibal (anchor, scuttle, Hela)	Not BREAK; soft escalate 0 survivors
Own-orders / filler $\times$ AAAA	Own: 0 physical; filler below soft — no rings
Catalog exact $\times$ right rings	Parked originalIndex 417/2513; <code>--bombe-from 418</code>
Middle ring $\neq$ A	Untested vs P1030680; arm graded end-to-end (control breaks at full middle $\times$ right coverage, $26\times$ space for $12.2\times$ time)
Long menus (len 28–40)	Prior negatives <i>incomplete</i> : old kernel cap silently eliminated up to 5.56% of lanes. Phase 16 arm 2 contaminated 40/40, Phase 8 top-30 20/30, catalog 300/2513; ≤ 25 letters unaffected. Re-runs under way
Ciphertext garble risk	Sister P1030681 needed degarbling; exact bars may drop true shells

Table 1: Campaign grades (cleartext Metal). Fitness is not HELUT encrypted tick rate (N6).

Soft-band Welchman stops can quarantine into Stochastic seeds (`--hybrid-quarantine`). Synthetic P1030684 block-wipe and historical P1030681 first-draft controls grade that path (escalate recovers edit ceilings, not a new key). On P1030680 itself, UEBUNG and Regenbogen soft bands produced German-looking IC without survivors under the 80% bar — coincidence, not plaintext.

6.1 A clean negative the board had not earned

The Metal bombe tabulates one upper-involution table per distinct slow-wheel (l, m) state across a menu span, and capped that at four. On overflow it wrote a zero survivor mask — the code for *eliminated* — for a lane it had never tested. Pure stepping arithmetic over every span, all 56 naval (middle, right) rotor pairs and all 676 windows (`Scripts/max_upper_audit.py`) bounds how often that fired: never at or below 25 letters, then 0.29 % of lanes at 28, 1.14 % at 30 and 5.56 % at 40. The damage is concentrated in the curated long-menu arms rather than spread thinly. Phase 16 arm 2 — forty mid-message anchors whose zero raw stops closed the Übung-push hypothesis — is 40 letters throughout, so *all forty* placements were contaminated and that negative was unearned as recorded. It has since been re-run on the corrected kernel: 40/40 dead at the board, zero raw stops over 6.387×10^{11} settings, so the cap was not concealing a key and the hypothesis stays closed on evidence. Phase 8’s curated top-30 is contaminated on 20 of 30 placements, and the main catalog on 300 of 2513. The Regenbogen queue is clean, and everything at 16–25 letters, including all of the Thetis-register work, is untouched and stands. The cap is now 8 and overflow reports *undecided*, which the host — which has no such cap — re-tests in full: a table size can no longer discard a key. Re-runs are ordered by contaminated share per GPU-hour, Phase 16 arm 2 first.

6.2 Unpinning the middle ring for 6.5–11.4×, not 26×

Every campaign to date pinned the middle Ringstellung at A, which is exact only while the middle wheel stays off its own notch in span. The gap is real (~ 8 –15 % of keys) and was priced at a flat 26×. It is cheaper, because notch tests depend on *window* position and are ring-free: if neither a lane’s middle wheel nor that of its ring-A partner at window $m - \rho$ reaches a notch across the span, the middle wheel steps only on right-wheel notches — the same times in both trails — the left wheel never moves, and $\text{offset}_M = m - \rho$ is the partner’s window. Same scramblers, same verdict. Ring A therefore runs in full and rings B–Z run only their notch-hitting lanes, and the union is complete coverage rather than a sample. Graded on the validated host board over 2.4×10^6 claimed-covered pairs and then through the kernel: zero verdict mismatches, and with $\sim 5\,600$ – $6\,900$ survivors per ring actually discarded, every discarded verdict is carried identically by its ring-A partner. Predicted cost is 6.5× a right-ring pass with a one-notch middle wheel and 11.4× with two. End-to-end on the blind control at full middle × right coverage — 908 544 shells, 4.152×10^{11} settings, the first run in this campaign with no residual ring gap — the break is recovered with all ten plugs in 4 307 s at 96.4 M/s: 26× the search space for 12.2× the time of the right-ring pass, because roughly 70 % of lanes exit after the trail test. Raw stops rise from 2 to 4, which is the mechanism working rather than a regression: with the middle ring unpinned the true machine has several ring-equivalent representations in the swept space. Measured basis is ~ 72 minutes per placement. Implemented and graded is not run: as of this revision the arm has produced no P1030680 verdict.

Two sieves were graded and rejected. The historical ten-lead board bounds the plugs a menu has not yet forced ($26 - \text{determined} \geq 2(10 - \text{pairs})$); it is sound — the blind control still breaks — and useless, taking 847 434 raw stops to 847 414 (0.002 %) for a 4 % throughput loss, because it only bites on long strong menus that already die at the board. Separately, three quarantine flags this report and the ledger both document were never parsed, so the soft-tail arm recorded at -4.3 ran at the default -4.0 .

7 Parallel Track: TensorLUT

Independently of the Boolean catalog, HELUT ships a continuous-to-discrete compiler: Yosys LUT6 cells become 64-wide floating INIT tensors, Metal evaluates a multilinear stream, λ squeezes toward binary, and an emitter writes gate-level Verilog. Unmutated `enigma_m4` locks

$F_{\text{crypto}} = 0$ (925 LUT6 + 49 DFFs baseline). Full INIT cold-start and stecker-cone melts *shatter* under λ ; the live arm freezes known-good silicon and evolves a ≤ 10 -pair stecker involution by construction (blind three-pair rediscovery PASS; claims C8–C9). Six formal lemmas of the continuous–discrete loop hold in-repo (claim C19; `directives/tensorlut-theorem.md`). Melt–freeze–snap on a separable Boolean interpolant is claim C44; a 2-LUT cascade melt that emits Verilog and matches 8 Boolean corners is claim C48 (dual interpolants allowed). This grades generative hardware synthesis and reciprocal genotype search. It does **not** invent P1030680’s missing plaintext (N7, H6).

8 Scope Relative to the HELUT Stack

HELUT also graduates a torus FHE path (LWE/GLWE, GGSW blind-rotate, certificates) documented in `paper/helut.tex` and `directives/claim-sheet.md`. That path is out of scope for campaign fitness: P1030680 search remains cleartext Metal batch. Encrypted SING envelopes and hardness calibration are stack claims (C4–C7, H1–H3), not evidence that this message is broken.

9 Conclusion

P1030680 remains unbroken. The contribution of this campaign is a graded, reproducible Boolean and stochastic laboratory on Apple Silicon; archival elimination of wrong nets and many wrong wedges; scoring and ghost-control fixes that make short-M4 search honest; and a parallel TensorLUT arm that learns reciprocal structure without claiming a decrypt. The next archival or combinatorial lever—not more GPU on a wrong template—is what would change the ledger.